

5G RAN

Standards Compliance Feature Parameter Description

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1 Change History

The following describes the specific changes in different issues.

5G RAN10.1 Draft B (2026-01-30)

This issue includes the following changes.

- Technical changes
None
- Editorial changes
Added descriptions of the dependency relationship between interfering cell detection and 3D coverage pattern. For details, see [11.3.2.1 Prerequisite Functions](#).
Added descriptions of the dependency relationship between the slot structure and 3D coverage pattern. For details, see [11.3.2.1 Prerequisite Functions](#).
Revised other descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

5G RAN10.1 Draft A (2025-12-31)

This issue introduces the following changes to 5G RAN9.1 04 (2025-08-30).

- Technical changes

Change Description	Parameter Change	RAT	Base Station Model
Added support for the Uplink Slot Configuration Optimization feature by LampSite base stations in high frequency bands. For details, see 12.3.3 Hardware .	None	High-frequency TDD	DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
Added the slot structure SS05080402 for the Uplink Slot Configuration Optimization feature. For details, see 12.1 Principles .	Modified parameter: Added SS05080402 to the value range of the NRDUCell.SlotStructure parameter.	High-frequency TDD	5900 series base stations (macro base stations) - DBS3900 LampSite and DBS5900 LampSite
Added the constraint relationship between the PRACH configuration index and the cell radius in the Uplink Slot Configuration Optimization feature. For details, see 12.3.4 Cells .	None	High-frequency TDD	5900 series base stations (macro base stations) - DBS3900 LampSite and DBS5900 LampSite
Added the Uplink Slot Configuration Optimization feature. For details, see 12 Uplink Slot Configuration Optimization (High-Frequency TDD) .	Modified parameter: Added SS221 to the value range of the NRDUCell.SlotStructure parameter.	High-frequency TDD	5900 series base stations (macro base stations)

- Editorial changes

Added descriptions of theoretical maximum cell radius under the slot structures **SS221** and **SS05080402** in the Uplink Slot Configuration Optimization feature. For details, see [12.3.4 Cells](#).

Revised other descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Features in This Document

This document describes the following features.

Feature ID	Feature Name	Chapter/Section
FBFD-010001	3GPP R15 Standards Compliance	4 3GPP R15 Standards Compliance
FBFD-051101	3GPP R16 Standards Compliance	5 3GPP R16 Standards Compliance
FBFD-070101	3GPP R17 Standards Compliance	6 3GPP R17 Standards Compliance
FBFD-081101	3GPP R18 Standards Compliance	7 3GPP R18 Standards Compliance
FBFD-010002	LDPC+Polar Codes	8 LDPC+Polar Codes
FBFD-010004	Basic Numerology	9 Basic Numerology
FBFD-010005	Self-contained Frame Structure	10 Self-contained Timeslot (TDD)
FBFD-020100	Slot Configuration	11 Slot Configuration (TDD)
FBFD-100100	Uplink Slot Configuration Optimization	12 Uplink Slot Configuration Optimization (High-Frequency TDD)
FBFD-010006	F-OFDM	13 F-OFDM

2.3 Differences

Differences Between NSA and SA

Function Name	Difference	Chapter/Section
3GPP R15 Standards Compliance	None	4 3GPP R15 Standards Compliance
3GPP R16 Standards Compliance	None	5 3GPP R16 Standards Compliance

Function Name	Difference	Chapter/Section
3GPP R17 Standards Compliance	None	6 3GPP R17 Standards Compliance
3GPP R18 Standards Compliance	None	7 3GPP R18 Standards Compliance
LDPC+Polar Codes	None	8 LDPC+Polar Codes
Basic Numerology	None	9 Basic Numerology
Self-contained Timeslot (TDD)	None	10 Self-contained Timeslot (TDD)
Slot Configuration (TDD)	None	11 Slot Configuration (TDD)
Uplink Slot Configuration Optimization	None	12 Uplink Slot Configuration Optimization (High-Frequency TDD)
F-OFDM	None	13 F-OFDM

Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

Function Name	Difference	Chapter/Section
3GPP R15 Standards Compliance	None	4 3GPP R15 Standards Compliance
3GPP R16 Standards Compliance	None	5 3GPP R16 Standards Compliance
3GPP R17 Standards Compliance	None	6 3GPP R17 Standards Compliance

Function Name	Difference	Chapter/Section
3GPP R18 Standards Compliance	None	7 3GPP R18 Standards Compliance
LDPC+Polar Codes	None	8 LDPC+Polar Codes
Basic Numerology	Supported in both high and low frequency bands, with the following differences: SCS configurations supported by high and low frequency bands differ.	9 Basic Numerology
Self-contained Timeslot (TDD)	None	10 Self-contained Timeslot (TDD)
Slot Configuration (TDD)	Supported in both high and low frequency bands, with the following differences: Slot configurations supported by high and low frequency bands differ. DL Blanking is supported only in low frequency bands.	11 Slot Configuration (TDD)
Uplink Slot Configuration Optimization	Supported only in high frequency bands	12 Uplink Slot Configuration Optimization (High-Frequency TDD)
F-OFDM	Supported in both high and low frequency bands, with the following differences: Spectrum utilization supported by high and low frequency bands differs.	13 F-OFDM

3 Overview

This document describes LDPC+Polar Codes, Basic Numerology, Self-contained Timeslot (TDD), Slot Configuration (TDD), and filtered orthogonal frequency division multiplexing (F-OFDM), which are basic functions defined in 3GPP Release 15. This document mainly provides the principles for these functions. Parameter settings are involved for Slot Configuration but not other functions. Therefore, operation and maintenance are described only for Slot Configuration.

4 3GPP R15 Standards Compliance

Huawei is an active participant and a major contributor to the development of 3GPP specifications. In addition, Huawei strictly complies with 3GPP specifications during product development. Huawei 5G RAN5.0 gNodeBs comply with 3GPP Release 15 (released in March 2020).

5 3GPP R16 Standards Compliance

Huawei is an active participant and a major contributor to the development of 3GPP specifications. In addition, Huawei strictly complies with 3GPP specifications during product development. Huawei gNodeBs in 5G RAN5.1 and later versions comply with 3GPP Release 16 (released in December 2020).

6 3GPP R17 Standards Compliance

Huawei is an active participant and a major contributor to the development of 3GPP specifications. In addition, Huawei strictly complies with 3GPP specifications during product development. Huawei gNodeBs in 5G RAN8.0 and later versions comply with 3GPP Release 17 (released in March 2023).

7 3GPP R18 Standards Compliance

Huawei is an active participant and a major contributor to the development of 3GPP specifications. In addition, Huawei strictly complies with 3GPP specifications during product development. Huawei gNodeBs in 5G RAN9.0 and later versions comply with 3GPP Release 18 (released in July 2024).

8 LDPC+Polar Codes

The low-density parity check (LDPC) code and polar code are new channel coding schemes used over the air interface. They were introduced in 3GPP Release 15. For details, see 3GPP TS 38.212 V15.5.0.

LDPC Code

LDPC codes, which are linear block codes, are used for 5G enhanced Mobile Broadband (eMBB) traffic channels. An LDPC code uses a parity check matrix for coding with the formula $HC^T = 0$, where C includes information bits and parity bits. Assuming that C_1 to C_5 are information bits and P_1 to P_5 are parity bits (P_1 to P_5 are obtained by encoding based on the parity check matrix H and information bits C_1 to C_5), if

$$C = (C_1, C_2, \dots, C_5, P_1, P_2, \dots, P_5)$$

$$H = \begin{pmatrix} 1 & 1 & 1 & 1 & 0 & 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 1 \end{pmatrix}$$

Then,

$$\begin{cases} C_1 + C_2 + C_3 + C_4 + P_1 + P_2 = 0 \\ C_1 + C_5 + P_2 + P_3 = 0 \\ C_2 + C_5 + P_1 + P_3 + P_4 = 0 \\ C_3 + P_4 + P_5 = 0 \\ C_4 + P_1 + P_5 = 0 \end{cases}$$

Table 8-1 provides a comparison between LDPC codes and turbo codes. (Turbo codes are used for LTE traffic channels.) A higher decoding speed results in a

higher peak data rate and lower power consumption. Therefore, LDPC codes better meet the data decoding requirements of 5G services than turbo codes.

Table 8-1 Decoding speed comparison between LDPC codes and turbo codes

Scenario	Speed Comparison
Low-bit-rate scenarios	Their decoding speeds are similar.
High-bit-rate scenarios	LDPC codes achieve a much higher decoding speed.

Polar Code

Polar codes, which are linear block codes, are used for 5G eMBB control channels. A polar code uses a coding matrix, and the basic matrix is as follows:

$$F = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$$

A Kronecker product operation is performed for a number of times, n , and then a polar code matrix with a length of $N = 2^n$ is generated. For example, if the Kronecker product operation is performed for three times, the resulting coding matrix is as follows:

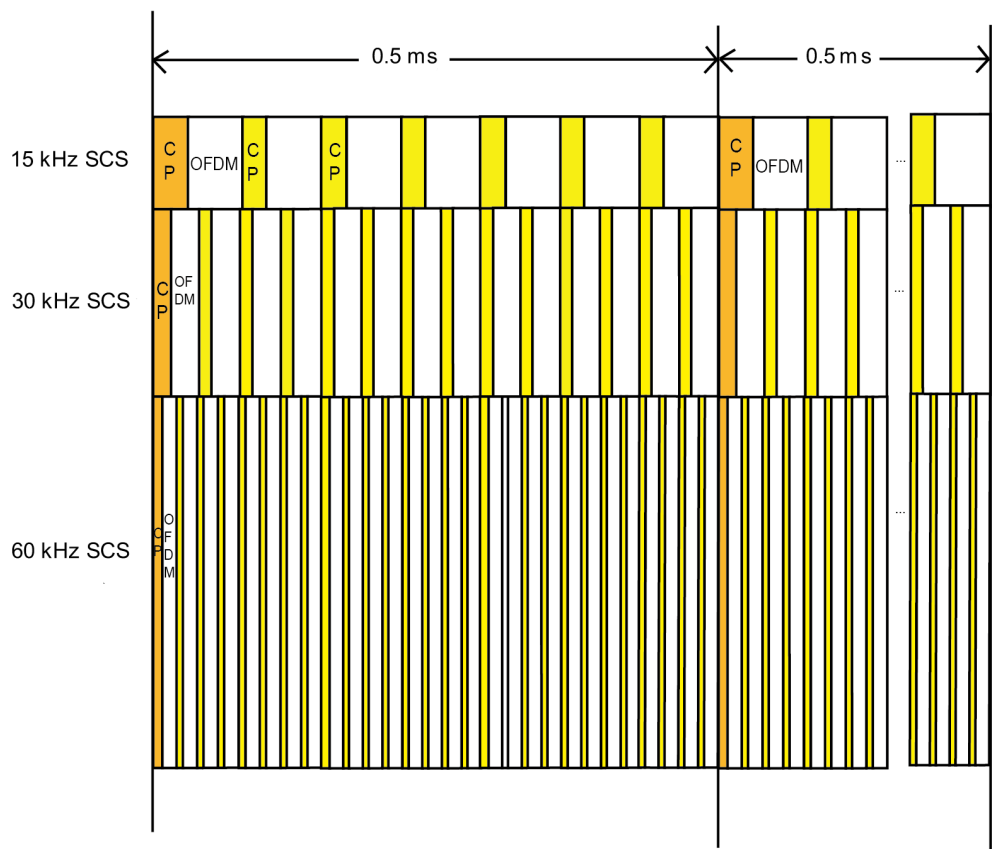
$$F^{\otimes 3} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

For a given block error rate (BLER), polar codes have a lower requirement for signal-to-noise ratio (SNR) than convolutional codes used for LTE control channels.

9 Basic Numerology

3GPP Release 15 has introduced basic numerologies. This release defines cyclic prefix (CP) lengths of different SCS configurations. For details, see 3GPP TS 38.211 V15.5.0. There are two types of CPs: normal and extended. An extended CP is supported only when the SCS is 60 kHz. [Figure 9-1](#) shows the symbols for SCSs 15 kHz, 30 kHz, and 60 kHz, with normal CP used as an example.

Figure 9-1 Basic numerologies



Within a period of 0.5 ms, CP lengths of different SCS configurations have the following characteristics:

- The CP length of the first symbol is greater than that of any other symbol.
- All CPs have the same length except for the CP of the first symbol.

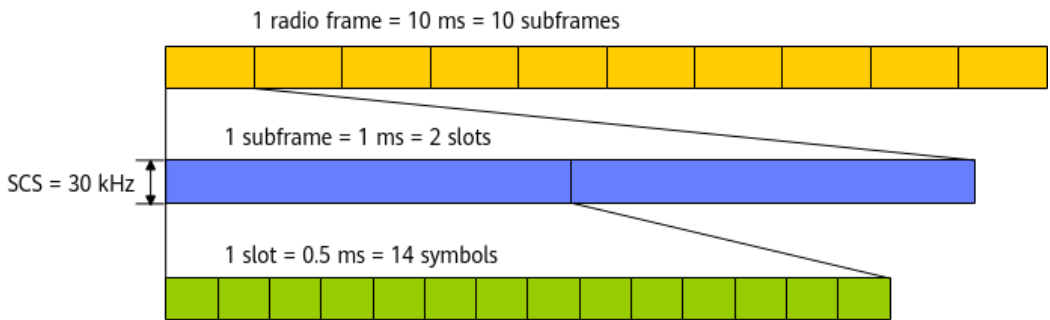
In NR TDD, basic numerologies currently support the 30 kHz and 120 kHz SCS of cells. [Table 9-1](#) lists the SCS configurations supported by different frequency bands, which are specified in 3GPP TS 38.104 V16.6.0.

Table 9-1 SCS configurations supported by different frequency bands

Frequency Band	Supported SCS
Sub-1 GHz	15 kHz and 30 kHz
1 GHz to 6 GHz	15 kHz, 30 kHz, and 60 kHz
24 GHz to 52.6 GHz	60 kHz and 120 kHz

The length of a subframe in NR is 1 ms, which is the same as that in LTE. As the SCS configuration varies depending on the frequency band, the number of slots in a subframe is changeable. For example, 15 kHz, 30 kHz, and 60 kHz correspond to one, two, and four slots, respectively. Each slot includes 14 symbols. [Figure 9-2](#) shows the frame structure with an SCS of 30 kHz.

Figure 9-2 Frame structure



Basic numerologies have the following characteristics in applications:

- Different SCS configurations can be used to satisfy various delay requirements for different services. A larger SCS results in shorter slots and shorter system delay.
- Moving UEs experience Doppler shifts. The faster a UE is moving, the larger the shift is. By increasing the SCS, the effect of this shift is reduced and the system is made more robust.
- A smaller SCS corresponds to a longer CP and a larger supported cell radius.

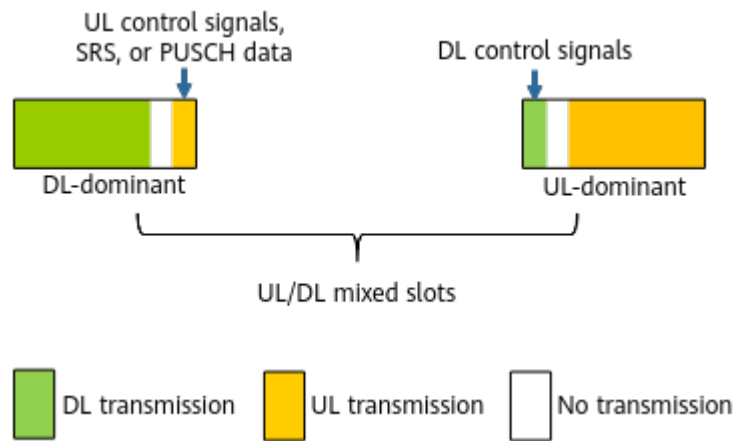
For details about the SCS configurations supported in the current version, see *Cell Management*.

10 Self-contained Timeslot (TDD)

The self-contained timeslot refers to the self-contained slot, which is a new slot format supported by 3GPP Release 15.

Generally, in TDD, uplink and downlink transmissions are separated into different slots. Uplink slots are used only for uplink transmission, and downlink slots are used only for downlink transmission. **A self-contained slot is a slot that supports both uplink and downlink transmissions.** Time division multiplexing enables uplink and downlink information to be transmitted during the same slot but using different orthogonal frequency division multiplexing (OFDM) symbols, as shown in [Figure 10-1](#).

Figure 10-1 Self-contained slot



There are two structures of self-contained slots, as described in [Table 10-1](#).

Table 10-1 Self-contained slot structures

Structure	Major Usage	Minor Usage	Advantage
DL-dominant slot	Downlink transmission	• Uplink control signal transmission • Sounding reference signal (SRS) transmission • Data transmission over the PUSCH	The delay of feedback for downlink transmission has been shortened.
UL-dominant slot	Uplink transmission	Downlink control signal transmission	The delay of uplink scheduling has been shortened.

 NOTE

Only DL-dominant slots are supported currently.

In self-contained slots, the switching between uplink and downlink transmissions must be performed on both the base station and UE sides. A guard period (the white part in [Figure 10-1](#)) is inserted between uplink and downlink transmissions in a self-contained slot to ensure proper signal transmission after the switching. No signal is transmitted or received within this period. Generally, the period is an integer multiple of the OFDM symbol length.

11 Slot Configuration (TDD)

11.1 Principles

11.1.1 General Principles

The following slot configurations are supported in low frequency bands with the SCS of 30 kHz:

- 4:1 slot configuration (4_1_DDDSU)
- Single-period 8:2 slot configuration (8_2_DDDDDDDSUU)
- Dual-period 8:2 slot configuration (8_2_DDDSUUDDDD)

The following slot configurations are supported in high frequency bands with the SCS of 120 kHz:

- 4:1 slot configuration (4_1_DDDSU)
- 2:3 slot configuration (2_3_DSUUU)

11.1.1.1 4:1 Slot Configuration

The 4:1 slot configuration (4_1_DDDSU) means a configuration of three DL-only slots (D), one self-contained slot (S), and one UL-only slot (U).

In low frequency bands where the SCS is 30 kHz, the 4:1 slot configuration (4_1_DDDSU) described in [Table 11-1](#) is supported. [Example of the 4:1 Slot Configuration \(4_1_DDDSU\)](#) provides an example.

In high frequency bands where the SCS is 120 kHz, the 4:1 slot configuration (4_1_DDDSU) described in [Table 11-2](#) is supported.

In the descriptions:

- X: indicates the configuration period.
High- and low-frequency cells differ in their subcarrier spacings and therefore use different configuration periods.
- x1: indicates the number of DL-only slots.

- y1: indicates the number of UL-only slots.
- x2: indicates the number of downlink symbols following DL-only slots.
- y2: indicates the number of uplink symbols before UL-only slots.
- GP: indicates the number of guard period (GP) symbols between uplink and downlink data transmission.

Table 11-1 Description of the 4:1 slot configuration (4_1_DDDSU) in low frequency bands

Slot Structure	GP	X (ms)	x1	x2	y2	y1
SS1	1	2.5	3	11	2	1
SS2	2	2.5	3	10	2	1
SS3	3	2.5	3	9	2	1
SS4	4	2.5	3	8	2	1
SS5	5	2.5	3	7	2	1
SS6	6	2.5	3	6	2	1
SS0100120 2	12	2.5	3	0	2	1
SS18	18	2.5	2	8	2	1

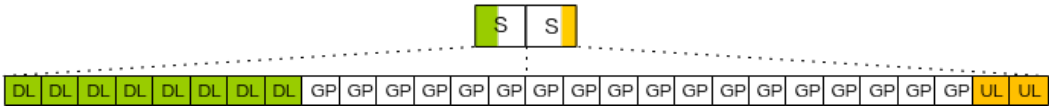
Table 11-2 Description of the 4:1 slot configuration (4_1_DDDSU) in high frequency bands

Slot Structure	GP	X (ms)	x1	x2	y2	y1
SS201	1	0.625	3	11	2	1
SS202	2	0.625	3	10	2	1
SS203	3	0.625	3	9	2	1
SS204	4	0.625	3	8	2	1
SS205	5	0.625	3	7	2	1
SS206	6	0.625	3	6	2	1

A slot contains 14 symbols. As such, the 18 GP symbols in a table, together with the downlink symbols preceding them and the uplink symbols following them, comprise 2 self-contained slots. In the example described in [Example of the 4:1 Slot Configuration \(4_1_DDDSU\)](#), when the slot structure is SS18, the number of GP symbols (GP) is 18, the number of downlink symbols (x2) is 8, and the number of uplink symbols (y2) is 2. These symbols comprise 2 slots: one consisting of 8

downlink symbols and 6 GP symbols, and the other consisting of 12 GP symbols and 2 uplink symbols, as shown in [Figure 11-1](#).

Figure 11-1 Slot structure with 18 GP symbols



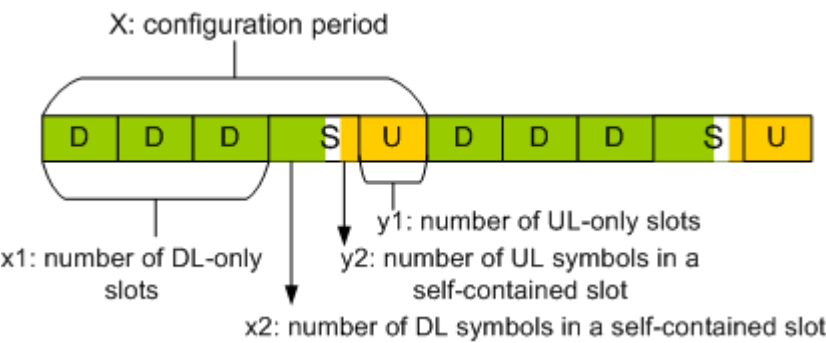
NOTE

For more details about self-contained slots, see [10 Self-contained Timeslot \(TDD\)](#).

Example of the 4:1 Slot Configuration (4_1_DDDSU)

If *NRDUCell.SlotAssignment* is set to 4_1_DDDSU and *NRDUCell.SlotStructure* is set to SS2, the values of parameters X, x1, x2, y2, and y1 for the SS2 slot structure are 2.5, 3, 10, 2, and 1, respectively. [Figure 11-2](#) shows the 4:1 slot configuration.

Figure 11-2 4:1 slot configuration (5 ms)



11.1.1.2 8:2 Slot Configuration

The single-period 8:2 slot configuration (8_2_DDDDDDDSUU) or dual-period 8:2 slot configuration (8_2_DDDSUDDDD) means a configuration of seven DL-only slots (D), one self-contained slot (S), and two UL-only slots (U).

In low frequency bands where the SCS is 30 kHz:

- The single-period 8:2 slot configuration (8_2_DDDDDDDSUU) described in [Table 11-3](#) is supported. [Example of the Single-Period 8:2 Slot Configuration \(8_2_DDDDDDDSUU\)](#) provides an example.
- The dual-period 8:2 slot configuration (8_2_DDDSUDDDD) described in [Table 11-4](#) and [Table 11-5](#) is supported. [Example of the Dual-Period 8:2 Slot Configuration \(8_2_DDDSUDDDD\)](#) provides an example.

In the descriptions:

- X or Y: indicates the configuration period.
High- and low-frequency cells differ in their subcarrier spacings and therefore use different configuration periods.

- x1 or x3: indicates the number of DL-only slots.
- y1 or y3: indicates the number of UL-only slots.
- x2 or x4: indicates the number of downlink symbols following DL-only slots.
- y2 or y4: indicates the number of uplink symbols before UL-only slots.
- GP: indicates the number of guard period (GP) symbols between uplink and downlink data transmission.

Table 11-3 Description of the single-period 8:2 slot configuration (8_2_DDDDDDSUU) in low frequency bands

Slot Structure	GP	X (ms)	x1	x2	y2	y1
SS51	1	5	7	9	4	2
SS52	2	5	7	8	4	2
SS53	3	5	7	7	4	2
SS54	4	5	7	6	4	2
SS55	5	5	7	5	4	2
SS56	6	5	7	4	4	2
SS518	18	5	6	6	4	2

Table 11-4 Description of the first-period slot configuration when the slot configuration is dual-period 8:2 (8_2_DDDSUUDDDD) in low frequency bands

Slot Structure	GP	X (ms)	x1	x2	y2	y1
SS81	1	3	3	9	4	2
SS82	2	3	3	8	4	2
SS83	3	3	3	7	4	2
SS84	4	3	3	6	4	2
SS85	5	3	3	5	4	2
SS86	6	3	3	4	4	2
SS818	18	3	2	6	4	2

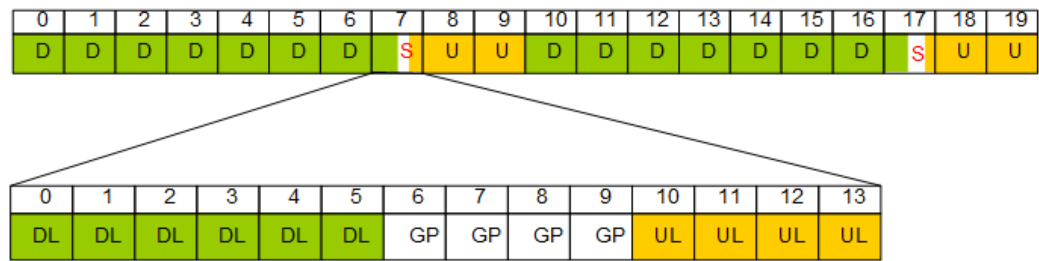
Table 11-5 Description of the second-period slot configuration when the slot configuration is dual-period 8:2 (8_2_DDDSUUDDDD) in low frequency bands

Slot Structure	GP	Y (ms)	x3	x4	y4	y3
SS81	1	2	4	0	0	0
SS82	2	2	4	0	0	0
SS83	3	2	4	0	0	0
SS84	4	2	4	0	0	0
SS85	5	2	4	0	0	0
SS86	6	2	4	0	0	0
SS818	18	2	4	0	0	0

Example of the Single-Period 8:2 Slot Configuration (8_2_DDDDDDDSUU)

If **NRDUCell.SlotAssignment** is set to **8_2_DDDDDDDSUU** and **NRDUCell.SlotStructure** is set to **SS54**, the configuration period is 5 ms, including 10 slots when the SCS is 30 kHz. **Figure 11-3** shows the 10 slots, including 7 DL-only slots (**D**), 1 self-contained slot (**S**), and 2 UL-only slots (**U**). The white area in **Figure 11-3** indicates the GP symbols between uplink and downlink data transmission. The self-contained slot includes 6 downlink symbols (**DL**), 4 GP symbols (**GP**), and 4 uplink symbols (**UL**).

Figure 11-3 Single-period 8:2 slot configuration (10 ms)



In an NR and LTE dual-mode network, the uplink reception and downlink transmission of one system must be synchronized with those of the other system, respectively, to ensure NR and LTE network synchronization. Otherwise, the downlink transmission of one system will interfere with the uplink reception of the other system. When **NRDUCell.SlotAssignment** is set to **8_2_DDDDDDDSUU**, parameter reconfiguration can be performed as follows to achieve network synchronization between NR and LTE TDD (with a subframe configuration of DSUDD), preventing adjacent-channel interference or interference caused by coexistence:

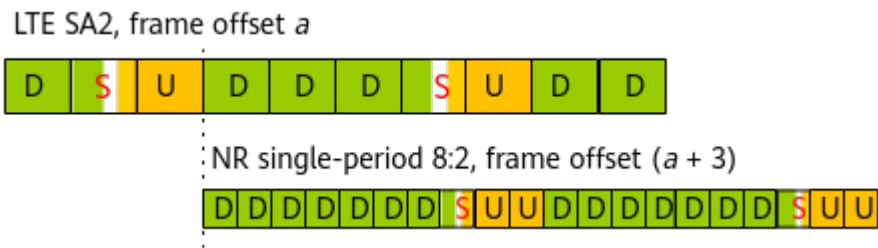
- Adjust the following NR parameters: **NRDUCell.SlotAssignment**, **NRDUCell.SlotStructure**, and **gNodeBParam.FrameOffset** or **gNBFreqBandConfig.FrameOffset**.

- Adjust the following LTE parameters: **Cell.SubframeAssignment**, **Cell.SpecialSubframePatterns**, and **ENodeBFrameOffset.TddFrameOffset** or **CellFrameOffset.FrameOffset**.

 NOTE

- Currently, the following compatibility between NR and LTE frame structures is supported. The effective value of the NR cell frame offset is 3 ms later than that of the LTE cell frame offset, as shown in [Figure 11-4](#).
- The NR single-period 8:2 slot configuration (8_2_DDDDDDDSUU) with slot structure SS54 is compatible with LTE uplink-downlink subframe configuration 2 (SA2) and special subframe configuration 7 (SSP7).
 - The NR single-period 8:2 slot configuration (8_2_DDDDDDDSUU) with slot structure SS56 is compatible with LTE uplink-downlink subframe configuration 2 (SA2) and special subframe configuration 6 (SSP6).
 - The NR single-period 8:2 slot configuration (8_2_DDDDDDDSUU) with slot structure SS518 is compatible with LTE uplink-downlink subframe configuration 2 (SA2) and special subframe configuration 5 (SSP5).

Figure 11-4 Frame structure for NR and LTE network synchronization

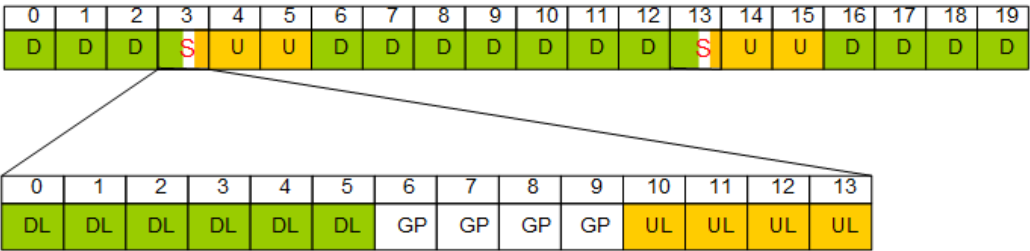


Example of the Dual-Period 8:2 Slot Configuration (8_2_DDDSUUDDDD)

- If **NRDUCell.SlotAssignment** is set to **8_2_DDDSUUDDDD** and **NRDUCell.SlotStructure** is set to **SS84**, the first configuration period is 3 ms and the second one is 2 ms, as shown in [Figure 11-5](#).
- The first period provides 3 DL-only slots (**D**), 1 self-contained slot (**S**), and 2 UL-only slots (**U**).
 - The second period provides 4 DL-only slots (**D**), 0 self-contained slots (**S**), and 0 UL-only slots (**U**).

The white area in [Figure 11-5](#) indicates the GP symbols between uplink and downlink data transmission. The self-contained slot includes 6 downlink symbols (**DL**), 4 GP symbols (**GP**), and 4 uplink symbols (**UL**).

Figure 11-5 Dual-period 8:2 slot configuration (10 ms)



In an NR and LTE dual-mode network, the uplink reception and downlink transmission of one system must be synchronized with those of the other system, respectively, to ensure NR and LTE network synchronization. Otherwise, the downlink transmission of one system will interfere with the uplink reception of the other system. When **NRDUCell.SlotAssignment** is set to **8_2_DDDSUUUUUUU**, data reconfiguration can be performed as follows to achieve network synchronization between NR and LTE TDD (with a subframe configuration of DSUDD), preventing adjacent-channel interference or interference caused by coexistence:

- Adjust the following NR parameters: **NRDUCell.SlotAssignment**, **NRDUCell.SlotStructure**, and **gNodeBParam.FrameOffset** or **gNBFreqBandConfig.FrameOffset**.
- Adjust the following LTE parameters: **Cell.SubframeAssignment**, **Cell.SpecialSubframePatterns**, and **ENodeBFrameOffset.TddFrameOffset** or **CellFrameOffset.FrameOffset**.

NOTE

In the current version, the following mapping between NR and LTE frame structures is supported. The effective value of the NR cell frame offset is aligned with that of the LTE cell frame offset, as shown in [Figure 11-6](#).

- NR dual-period 8:2 slot configuration (8_2_DDDSUUUUUUU) and slot structure SS84 map to LTE uplink-downlink subframe configuration SA2 and special subframe configuration SSP7.
- NR dual-period 8:2 slot configuration (8_2_DDDSUUUUUUU) and slot structure SS86 map to LTE uplink-downlink subframe configuration SA2 and special subframe configuration SSP6.
- NR dual-period 8:2 slot configuration (8_2_DDDSUUUUUUU) and slot structure SS818 map to LTE uplink-downlink subframe configuration SA2 and special subframe configuration SSP5.

Figure 11-6 Frame structure for NR and LTE network synchronization

LTE SA2, frame offset *a*



NR dual-period 8:2, frame offset *a*



11.1.2 DL Blanking

In the case that operators configure different slot configurations for intra-band cells in a boundary area, the downlink symbols (*DL*) of one slot configuration interfere with the uplink symbols (*UL*) or GP symbols (*GP*) of the other. Another case is that, when an operator evolves its LTE network into an NSA network and uses the single-period 8:2 slot configuration for frame synchronization between NR and LTE systems but another NR operator uses the 4:1 slot configuration for shorter latency in a boundary area, interference occurs between them. To address the interference in these cases, Huawei introduces the DL Blanking function. With this function, the base station does not perform scheduling in specified downlink symbols when different slot configurations are used.

 NOTE

The single-period 8:2 slot configuration (8_2_DDDDDDDSUU) requires a 3 ms larger frame offset than the 4:1 slot configuration (4_1_DDDSU) does, to ensure frame synchronization between NR and LTE systems or between different slot configurations within an NR system. The synchronization helps reduce the number of symbols with interference and wastes of transmission resources. For details, see [Figure 11-4](#) and [Figure 11-7](#).

Currently, DL Blanking works between the 4:1 slot configuration (4_1_DDDSU) and the single-period 8:2 slot configuration (8_2_DDDDDDDSUU). This function is supported only in low frequency bands n77 and n78.

This function is enabled by setting the **NRDUCell.PeerCellSlotAssignment** parameter to **4_1_DDDSU** or **8_2_DDDDDDDSUU**. The value of this parameter must be the same as the slot configuration of the peer cell, as described in [Table 11-6](#).

Table 11-6 Parameter settings for enabling the DL Blanking function

Peer Cell Slot Configuration	Local Cell Slot Configuration	Parameter Setting
4:1 slot configuration (4_1_DDDSU)	Single-period 8:2 slot configuration (8_2_DDDDDDDSUU)	4_1_DDDSU
Single-period 8:2 slot configuration (8_2_DDDDDDDSUU)	4:1 slot configuration (4_1_DDDSU)	8_2_DDDDDDDSUU

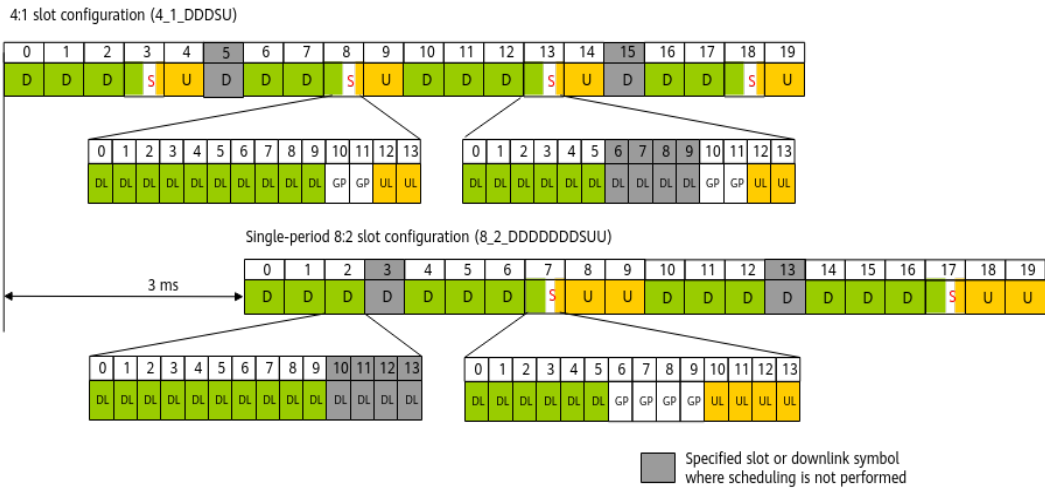
The principles for the DL Blanking function are detailed as follows:

- DL Blanking for the 4:1 slot configuration (4_1_DDDSU)
 - When the slot structure is SS1 to SS6, scheduling is specified to be not performed in slot 5, slot 15, downlink symbols among symbols 6 to 9 of slot 3, or downlink symbols among symbols 6 to 9 of slot 13. In this case, among the downlink symbols of slots 3 and 13, only symbols 0 to 5 are available for scheduling.
 - When the slot structure is SS01001202 or SS18, scheduling is only specified to be not performed in slot 5 or 15. Symbols 6 to 9 of slots 3 and 13 are GP symbols, in which scheduling is not performed either.
- DL Blanking for the single-period 8:2 slot configuration (8_2_DDDDDDDSUU)

Scheduling is specified to be not performed in slot 3, slot 13, downlink symbols 10 to 13 of slot 2, or downlink symbols 10 to 13 of slot 12. In this case, among the downlink symbols of slots 2 and 12, only symbols 0 to 9 are available for scheduling.

Take the SS2 slot structure of the 4:1 slot configuration and the SS54 slot structure of the single-period 8:2 slot configuration as examples. [Figure 11-7](#) shows how the DL Blanking function works in this example.

Figure 11-7 Example of DL Blanking (10 ms)



11.1.3 Interfering Cell Detection

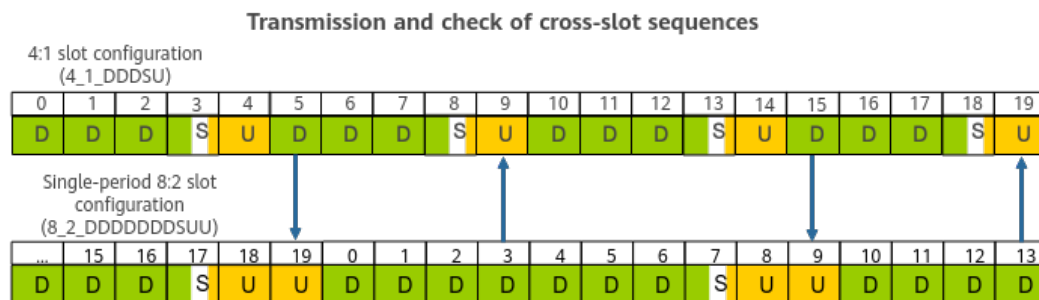
If different operators have configured the 4:1 slot configuration (4_1_DDDSU) and the single-period 8:2 slot configuration (8_2_DDDDDDDSUU) separately for intra-band cells in a boundary area, DL Blanking can be used. This function requires interfering cells to be identified. Interfering cells with the 4:1 slot configuration (4_1_DDDSU) and the single-period 8:2 slot configuration (8_2_DDDDDDDSUU) can be identified by sending and checking cross-slot sequences between cells with these slot configurations. For details about cross-slot sequences, see [Cross-Slot Sequence](#).

The interfering cell detection function is enabled by setting the **NRDUCellTimeIntrf.CrossLinkIntrfCfg** parameter to **CROSS_LINK_SEQ_S_D_TYPE1**. It is compatible only with low-frequency TDD cells in bands n77 and n78. As illustrated in [Figure 11-8](#), interfering cell detection works as follows:

- A cell with the 4:1 slot configuration (4_1_DDDSU) sends cross-slot sequences in slots 5 and 15 and checks cross-slot sequences in slots 9 and 19.
- A cell with the single-period 8:2 slot configuration (8_2_DDDDDDDSUU) sends cross-slot sequences in slots 3 and 13 and checks cross-slot sequences in slots 9 and 19.

The interval at which a cell sends cross-slot sequences is relevant to the setting of the **NRDUCellRimConfig.RimRsType1SetIdSize** parameter. This parameter determines the maximum number of interfering cells that can be detected. The smaller the parameter value, the shorter the interval. The larger the parameter value, the longer the interval.

Figure 11-8 Interfering cell detection



NOTE

- For interfering cell detection to take effect, this function must be enabled for all interfering and victim cells (that is, all intra-band cells with the slot configuration 4_1_DDDSU and the slot configuration 8_2_DDDDDDSUU in the boundary area) and the serving base stations must be Huawei products.
- When victim cells are highly loaded in the uplink, missing or false detection of cross-slot sequences may occur. To avoid this, it is recommended that interfering cell detection be performed during light-load periods.
- When there are multiple interfering cells, missing detection of cross-slot sequences may occur. To avoid this, it is recommended that interfering cell detection be performed continuously after DL Blanking is deployed for each detected interfering cell.

Cross-Slot Sequence

The cross-slot sequences used in this function are remote interference management reference signals (RIM-RSs) generated using type-1 RIM-RS configurations. For the definition of RIM-RS and details about the generation process, see *Remote Interference Management (Low-Frequency TDD)*. To distinguish the RIM-RSs used for interfering cell detection from those used for remote interference management, different formulas are used in the two functions to calculate the start frequency-domain position of RIM-RSs for a cell. In addition, different methods are used in the two functions to determine how to set the **NRDUCellRimConfig.RimRsStartFreqOffset** and **NRDUCellRimConfig.RimRsStartFreqReOffset** parameters, which are involved in the calculation of the start position.

For interfering cell detection, the start frequency-domain position of RIM-RSs (that is, cross-slot sequences to be sent) of a cell is calculated using the following formula:

RIM-RS start frequency-domain position of a cell = Point A of the cell + **NRDUCellRimConfig.RimRsStartFreqOffset** + **NRDUCellRimConfig.RimRsStartFreqReOffset** + *OFFSET*

- Point A of the cell: For the definition of point A, see 3GPP TS 38.211 V16.5.0.
- **NRDUCellRimConfig.RimRsStartFreqOffset**: It specifies the offset (in units of RBs) for the start frequency-domain position of RIM-RSs. The reference point is point A.
- **NRDUCellRimConfig.RimRsStartFreqReOffset**: It specifies the offset (in units of REs) for the start frequency-domain position of RIM-RSs. The reference point is point A + **NRDUCellRimConfig.RimRsStartFreqOffset**.

- **OFFSET**: It denotes the offset (in units of REs) of RIM-RSs for this function relative to RIM-RSs for remote interference management. The gNodeB automatically generates the value of **OFFSET** according to the following rule: If the result of $(\text{NRDUCellRimConfig.RimRsStartFreqOffset} \times 12 + \text{NRDUCellRimConfig.RimRsStartFreqReOffset})$ is less than or equal to the reference value for **OFFSET** evaluation under the specified combination of the number of candidate frequency-domain resources for RIM-RSs and the cell bandwidth, then **OFFSET** is equal to 6. Otherwise, **OFFSET** is equal to -6. [Table 11-7](#) presents the reference values.

Table 11-7 Reference values for **OFFSET** evaluation under various combinations of the number of candidate frequency-domain resources for RIM-RSs and the cell bandwidth

Cell Bandwidth	Value Under One Candidate Frequency-Domain Resource for RIM-RSs ^a	Value Under Two Candidate Frequency-Domain Resources for RIM-RSs ^b	Value Under Four Candidate Frequency-Domain Resources for RIM-RSs ^c
20 MHz	18	/	/
30 MHz	180	/	/
40 MHz	348	60	/
50 MHz	510	222	/
60 MHz	684	396	/
70 MHz	846	558	/
80 MHz	1014	726	150
90 MHz	1182	894	318
100 MHz	1350	1062	486
• a: The NRDUCellRimConfig.RimRsFreqCandNum parameter is set to FREQ_NUM_1, indicating that there is only one candidate frequency-domain resource for RIM-RSs. • b: The NRDUCellRimConfig.RimRsFreqCandNum parameter is set to FREQ_NUM_2, indicating that there are two candidate frequency-domain resources for RIM-RSs. • c: The NRDUCellRimConfig.RimRsFreqCandNum parameter is set to FREQ_NUM_4, indicating that there are four candidate frequency-domain resources for RIM-RSs. • Note: Slashes (/) in the table indicate that there is no reference value for OFFSET evaluation under the specified combination of the number of candidate frequency-domain resources for RIM-RSs and the cell bandwidth.			

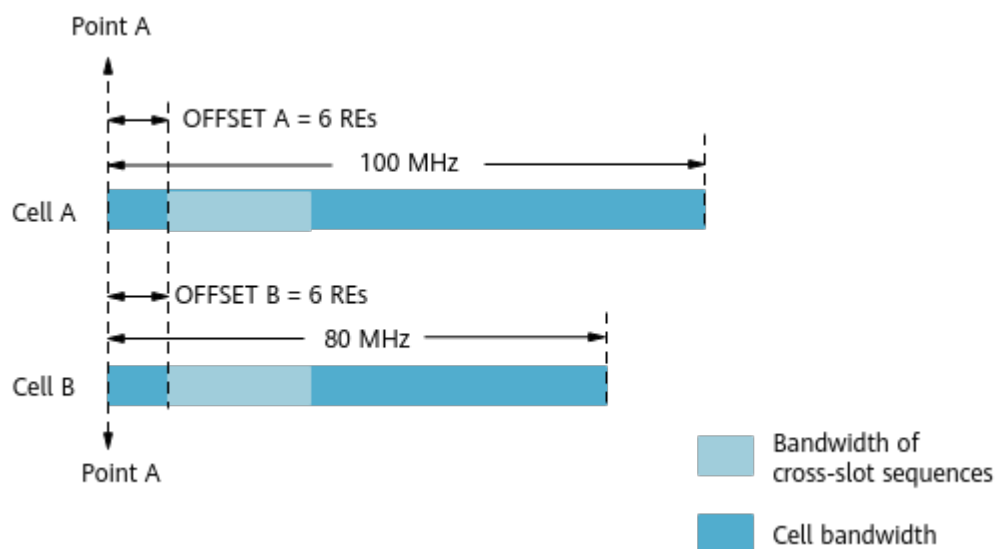
Parameters for Start Frequency-Domain Positions of RIM-RSs in Cells

As RIM-RSs span 20 MHz (48 RBs) in the frequency domain, it is required that the bandwidth of the interfering cell and that of the victim cell (engaged in mutual RIM-RS transmission and check) overlap by more than 20 MHz and the subcarrier spacings of the two cells be the same. There are different scenarios regarding the start frequency-domain positions in the interfering and victim cells. The positions may be aligned and non-aligned. When they are non-aligned, their difference may be small or large. The method of setting the **NRDUCellRimConfig.RimRsStartFreqOffset** and **NRDUCellRimConfig.RimRsStartFreqReOffset** parameters, which determine the start frequency-domain positions of RIM-RSs, varies depending on the scenarios.

Aligned Start Frequency-Domain Positions in the Interfering and Victim Cells

As shown in [Figure 11-9](#), cells A and B belong to different operators. The start frequency-domain positions in the cells are aligned. The two cells are interfering and victim cells of each other. In this scenario, set the **NRDUCellRimConfig.RimRsStartFreqOffset** and **NRDUCellRimConfig.RimRsStartFreqReOffset** parameters to their default value 0 for the two cells, and *OFFSET* is equal to 6 for both cells. In this case, the positions of the cross-slot sequences in the interfering and victim cells are the same, with the start frequency-domain positions being (point A + 6 REs) in both cells.

Figure 11-9 Frequency-domain positions of cross-slot sequences with the start positions aligned between cells



Non-aligned Start Frequency-Domain Positions in the Interfering and Victim Cells

If the interfering and victim cells are not aligned in their start frequency-domain positions, the cross-slot sequences in the two cells must be placed at the same positions by setting the **NRDUCellRimConfig.RimRsStartFreqOffset** and **NRDUCellRimConfig.RimRsStartFreqReOffset** parameters. The method of setting these parameters varies depending on the following cell conditions:

- Scenario 1: There is a small difference between the start frequency-domain positions of the two cells.

Cells A and B shown in **Figure 11-10** belong to different operators, and their frequency-domain information is listed in **Table 11-8**. According to *Remote Interference Management (Low-Frequency TDD)*, taking point A of cell B as a reference, the **NRDUCellRimConfig.RimRsStartFreqOffset** and **NRDUCellRimConfig.RimRsStartFreqReOffset** parameters for cell B should be both set to **0**. In addition, as there is a small difference (3 REs) between point A of cell A and point A of cell B, the **NRDUCellRimConfig.RimRsStartFreqOffset** and **NRDUCellRimConfig.RimRsStartFreqReOffset** parameters for cell A should be set to **0** and **3**, respectively. As the overlapping bandwidth between cells A and B is 80 MHz, the **NRDUCellRimConfig.RimRsFreqCandNum** parameter should be set to **FREQ_NUM_4** for both cells. Based on these parameter settings and **Table 11-7**, **OFFSET** is 6 for cell A (because " $0 \times 12 + 3 < 486$ ") and is also 6 for cell B (because " $0 \times 12 + 0 < 150$ "). In this case, the start frequency-domain position for cross-slot sequences is (Point A of cell A + $0 \times 12 + 3$ REs + 6 REs) for cell A and is (Point A of cell B + $0 \times 12 + 0 + 6$ REs) for cell B. This way, the cross-slot sequences are placed at the same positions in the interfering and victim cells.

Figure 11-10 Frequency-domain positions of cross-slot sequences with the start positions non-aligned between cells - scenario 1

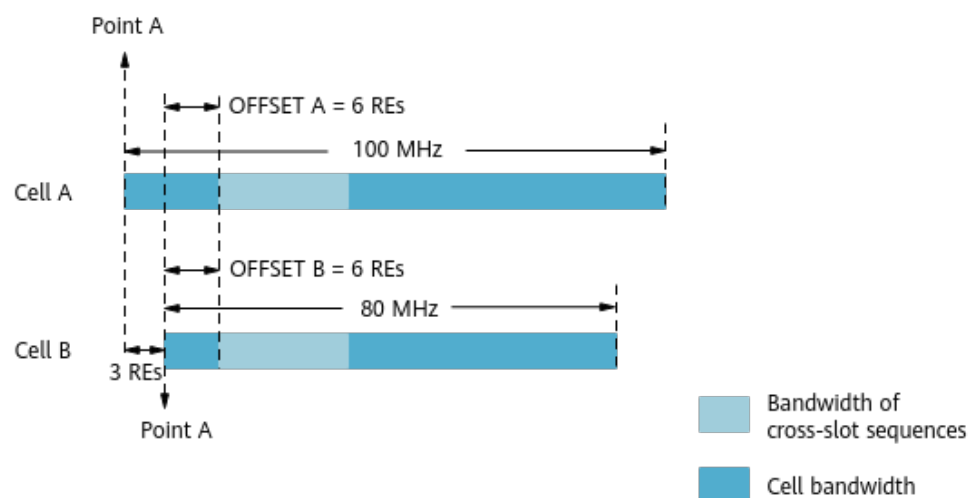


Table 11-8 Frequency-domain information for scenario 1

Cell	Band	Cell Center Frequency Number	Frequency of Point A	Subcarrier Spacing	Frequency Difference	Number of RBs and REs Corresponding to the Frequency Difference
A	3500–3600 MHz	636666	3500.85 MHz	30 kHz	90 kHz	0 RB + 3 REs
B	3500–3580 MHz	636000	3500.94 MHz	30 kHz	0 (cell B as a reference)	0 RB + 0 RE

- Scenario 2: There is a large difference between the start frequency-domain positions of the two cells.

Cells A and B shown in [Figure 11-11](#) belong to different operators, and their frequency-domain information is listed in [Table 11-9](#). According to *Remote Interference Management (Low-Frequency TDD)*, taking point A of cell B as a reference, the **NRDUCellRimConfig.RimRsStartFreqOffset** and **NRDUCellRimConfig.RimRsStartFreqReOffset** parameters for cell B should be both set to **0**. In addition, as there is a large difference (166 RBs + 8 REs) between point A of cell A and point A of cell B, the **NRDUCellRimConfig.RimRsStartFreqOffset** and **NRDUCellRimConfig.RimRsStartFreqReOffset** parameters for cell A should be set to **166** and **8**, respectively. As the overlapping bandwidth between cells A and B ranges from 20 MHz to 40 MHz, the **NRDUCellRimConfig.RimRsFreqCandNum** parameter should be set to **FREQ_NUM_1** for both cells. Based on these parameter settings and [Table 11-7](#), **OFFSET** is **–6** for cell A (because " $166 \times 12 + 8 > 1014$ ") and is **6** for cell B (because " $0 \times 0 + 0 < 150$ "). In this case, the start frequency-domain position for cross-slot sequences is (Point A of cell A + 166 RBs + 8 REs – 6 REs) for cell A and is (Point A of cell B + 6 REs) for cell B. This way, the cross-slot sequences are placed at different positions (with a 12-RE difference) in the interfering and victim cells, as shown in the upper part of [Figure 11-11](#).

To place the cross-slot sequences at the same positions in the two cells, the parameter settings for cell A must be adjusted so that the start frequency-domain position for cross-slot sequences in cell A (Point A + **NRDUCellRimConfig.RimRsStartFreqOffset** × 12 + **NRDUCellRimConfig.RimRsStartFreqReOffset** – 6 REs) is equal to that in cell B (Point A of cell B + 6 REs, that is, Point A of cell A + 166 RBs + 8 REs + 6 REs). To this end, the **NRDUCellRimConfig.RimRsStartFreqOffset** and **NRDUCellRimConfig.RimRsStartFreqReOffset** parameters for cell A should be set to **167** and **8**, respectively. As a result, the cross-slot sequences are aligned between the cells, as shown in the lower part of [Figure 11-11](#).

Figure 11-11 Frequency-domain positions of cross-slot sequences with the start positions non-aligned between cells - scenario 2

There is a 12-RE difference in cross-slot sequence positions between cells A and B.

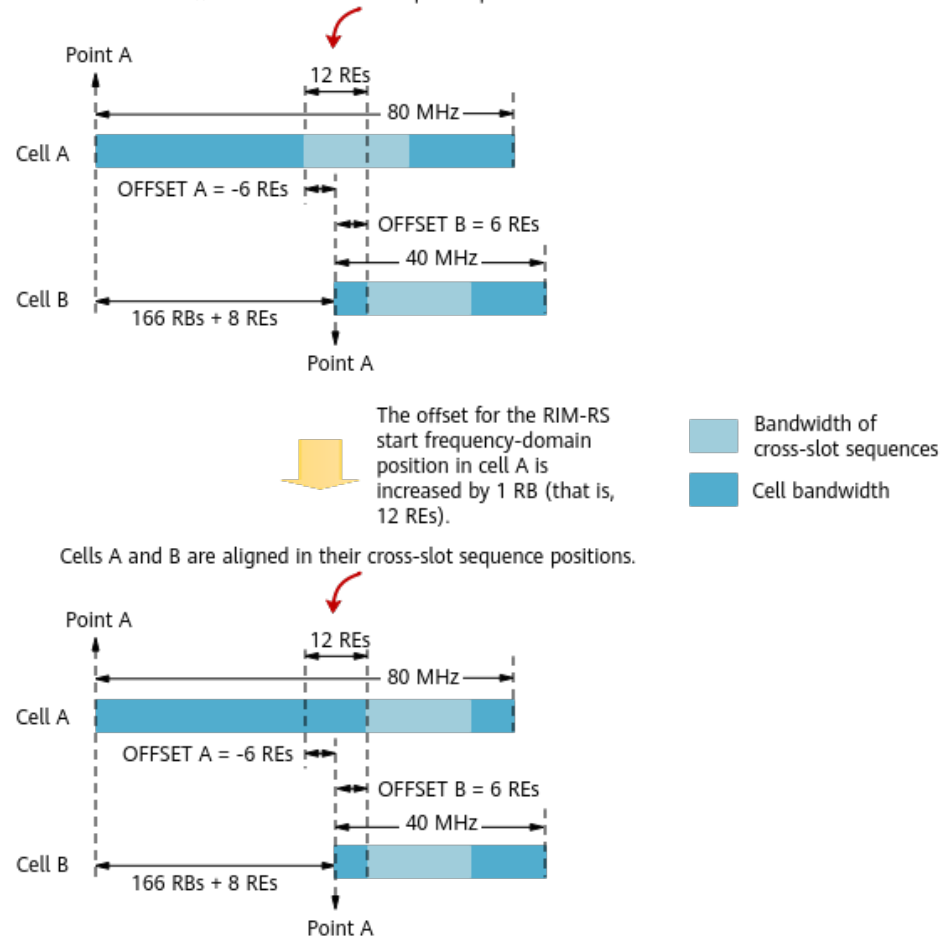


Table 11-9 Frequency-domain information for scenario 2

Cell	Band	Cell Center Frequency Number	Frequency of Point A	Subcarrier Spacing	Frequency Difference	Number of RBs and REs Corresponding to the Frequency Difference
A	3500–3580 MHz	636000	3500.94 MHz	30 kHz	60000 kHz	166 RBs + 8 REs
B	3560–3600 MHz	638668	3560.94 MHz	30 kHz	0 (cell B as a reference)	0 RB + 0 RE

11.2 Network Analysis

11.2.1 Benefits

DL Blanking prevents interference to uplink transmission when the 4:1 slot configuration (4_1_DDDSU) and the single-period 8:2 slot configuration (8_2_DDDDDDDSUU) are applied to intra-band cells in a boundary area.

Interfering cell detection resolves the issue of failure to identify interfering cells when DL Blanking is deployed.

11.2.2 Impacts

The impacts between this function and other functions are described in [11.3.2.3 Function Impacts](#).

DL Blanking

It is recommended that DL Blanking be used in boundary areas upon successful negotiation between operators at the initial stage of network construction. This function has the following impacts on the network:

- The downlink peak rate decreases because scheduling is not performed in specified downlink symbols of the 4:1 slot configuration (4_1_DDDSU) and the single-period 8:2 slot configuration (8_2_DDDDDDDSUU).
- The maximum number of SSBs decreases, and the SSB coverage shrinks.
- The total number of scheduled UEs in the downlink in a cell (indicated by the **N.User.Schedule.DL.Sum** counter) decreases.
- The total transmission delay of the first downlink RLC SDUs for DRB services in the empty buffer in a cell (indicated by the **N.Traffic.DL.RlcFirstPktDelay.Time** counter) increases.
- The duration of downlink data transmission at the MAC layer in a cell (indicated by the **N.ThpTime.DL.Cell** counter) decreases. When a cell is lightly loaded, the average downlink cell throughput (calculated using the formula: $1000 \times \text{N.ThpVol.DL.Cell} / \text{N.ThpTime.DL.Cell}$) increases, and the average uplink cell throughput remains unchanged.
- When the local and the peer cells are configured with different slot configurations (one with the 4:1 slot configuration of 4_1_DDDSU and the other with the single-period 8:2 slot configuration of 8_2_DDDDDDDSUU), their channel state information reference signals (CSI-RSs) are not aligned in the time domain. In this case, the CSI-RS of one cell interferes with the physical downlink shared channel (PDSCH) of the other.
- Assume that DL Blanking is enabled for a cell with the 4:1 slot configuration (4_1_DDDSU) and disabled for another cell with the 4:1 slot configuration (4_1_DDDSU). The tracking reference signal (TRS) transmission in slot 7 in the former cell interferes with two PDSCH symbols of slot 7 in the latter. Interference occurs only between adjacent cells, and fades to disappear as the distance between cells increases.
- Assume that DL Blanking is enabled for cell A with the 4:1 slot configuration (4_1_DDDSU) and cell B with the single-period 8:2 slot configuration (8_2_DDDDDDDSUU). The TRS transmission in slots 6 and 7 in cell A interferes with two PDSCH symbols of slot 0 and two PDSCH symbols of slot 1 in cell B. The TRS transmission in slots 5 and 6 in cell B interferes with two PDSCH symbols of slot 11 and two PDSCH symbols of slot 12 in cell A.

Interference occurs only between adjacent cells, and fades to disappear as the distance between cells increases.

Interfering Cell Detection

If operators have configured the 4:1 slot configuration (DDDSU) and the single-period 8:2 slot configuration (DDDDDDDSUU) for intra-band cells in a boundary area, it is recommended that interfering cell detection be enabled before DL Blanking is deployed. With this function enabled, interfering cells stop data scheduling when they send cross-slot sequences. This may cause the downlink peak rates of the cells to decrease.

11.3 Requirements

11.3.1 Licenses

None

11.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

11.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	3D coverage pattern	NRDUCellTrpBeam.CoverageScenario	<i>Beam Management</i>	When the NRDUCell.SlotStructure parameter is set to SS01001202 , SS18 , SS518 , and SS818 , the coverage scenario can only be set to: • DEFAULT • SCENARIO_1 to SCENARIO_16 • EXPAND_SCENARIO_7 • EXPAND_SCENARIO_8 • SCENARIO_27 • SCENARIO_28 • SCENARIO_29 • SCENARIO_30 • SCENARIO_32 • SCENARIO_33 • SCENARIO_34 • SCENARIO_35 • SCENARIO_37 • CUSTOMIZED_BEAM_SCENARIO

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	3D coverage pattern	NRDUCellTrpBeam.CoverageScenario	<i>Beam Management</i>	When interfering cell detection is enabled (with NRDUCellTimeInterf.CrossLinkInterf.Cfg set to CROSS_LINK_SEQ_S_D_TYPE1), the coverage scenario can only be set to DEFAULT , SCENARIO_1 to SCENARIO_16 , SCENARIO_36 , and CUSTOMIZED_BEAM_SCENARIO .

11.3.2.2 Mutually Exclusive Functions

Some features or functions are not supported when certain slot configurations are used. For details about the slot configuration requirements of features or functions, see the cell requirements described in the corresponding feature documents. The following table describes the functions that are subjected to slot configuration constraints but are not conveyed in any feature documents.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Deployment Position	NRDUCell.DeployPosition	None	When Deployment Position is set to DEPLOY_IAB: • If NRDUCell.SlotAssignment is set to 8_2_DDDDDDDSUU, NRDUCell.SlotStructure cannot be set to SS51 or SS518. • DL Blanking cannot be enabled. That is, NRDUCell.PeerCellSlotAssignment can only be set to NOT_CONFIG.
Low-frequency TDD	MT access control	MT_RRC_CO NN_SWITCH option of the NRDUCellConnMgmt.labCtrlSwitch parameter	None	When the MT access control switch is turned on and NRDUCell.SlotAssignment is set to 8_2_DDDDDDDSUU, NRDUCell.SlotStructure cannot be set to SS51.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink symbol interference avoidance	DL_SYMBOL_INTRF_AVOID_SW option of the NRDUCellTimeIntrfAvoidance parameter	None	When the downlink symbol interference avoidance switch is turned on, • If NRDUCell.SlotAssignment is set to 4_1_DDDSU, NRDUCell.SlotStructure cannot be set to SS01001202. • DL Blanking cannot be enabled. That is, NRDUCell.PeerCellSlotAssignment can only be set to NOT_CONFIG.
Low-frequency TDD	PDCCH symbol optimization	PDCCH_SYMBOL_OPT_SW option of the NRDUCellMntResourceOptSw parameter	None	When the PDCCH symbol optimization switch is turned on, the NRDUCell.SlotStructure parameter cannot be set to SS01001202 .

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Enhanced intra-frequency SSB interference avoidance	INTRA_FREQ_SSB_AVOID_ENH_SW option of the NRDUCellAlgoSwitch.CommChnIntrfAvoidSwitch parameter	<i>3D Networking Experience Improvement</i>	If the NRDUCell.PeerCellSlotAssignment parameter is set to a value other than NOT_CONFIG , the INTRA_FREQ_SSB_AVOID_ENH_SW option of the NRDUCellAlgoSwitch.CommChnIntrfAvoidSwitch parameter must be deselected.
Low-frequency TDD	Inter-frequency SSB interference avoidance	INTER_FREQ_SSB_AVOID_SW option of the NRDUCellAlgoSwitch.CommChnIntrfAvoidSwitch parameter	<i>3D Networking Experience Improvement</i>	When DL Blanking is enabled (by setting the NRDUCell.PeerCellSlotAssignment parameter to a value other than NOT_CONFIG), inter-frequency SSB interference avoidance cannot be enabled.
Low-frequency TDD	Enhanced inter-frequency SSB interference avoidance	INTER_FREQ_SSB_AVOID_ENH_SW option of the NRDUCellAlgoSwitch.CommChnIntrfAvoidSwitch parameter	<i>3D Networking Experience Improvement</i>	If the NRDUCell.PeerCellSlotAssignment parameter is set to a value other than NOT_CONFIG , the INTER_FREQ_SSB_AVOID_ENH_SW option of the NRDUCellAlgoSwitch.CommChnIntrfAvoidSwitch parameter must be deselected.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intra-frequency CSI-RS interference avoidance	INTRA_FREQ_CSIRS_AVOID_SW option of the NRDUCellAlgoSwitch.CommChnIntrfAvoidSwitch parameter	<i>3D Networking Experience Improvement</i>	If the NRDUCell.PeerCellsSlotAssignment parameter is set to a value other than NOT_CONFIG , the INTRA_FREQ_CSIRS_AVOID_SW option of the NRDUCellAlgoSwitch.CommChnIntrfAvoidSwitch parameter must be deselected.
Low-frequency TDD	Inter-frequency CSI-RS interference avoidance	INTER_FREQ_CSIRS_AVOID_SW option of the NRDUCellAlgoSwitch.CommChnIntrfAvoidSwitch parameter	<i>3D Networking Experience Improvement</i>	If the NRDUCell.PeerCellsSlotAssignment parameter is set to a value other than NOT_CONFIG , the INTER_FREQ_CSIRS_AVOID_SW option of the NRDUCellAlgoSwitch.CommChnIntrfAvoidSwitch parameter must be deselected.
Low-frequency TDD	Common channel interference avoidance	SSB_INTRF_AVOID_SW option of the NRDUCellAlgoSwitch.CommChnIntrfAvoidSwitch parameter	<i>Interference Avoidance</i>	When DL Blanking is enabled (by setting the NRDUCell.PeerCellsSlotAssignment parameter to a value other than NOT_CONFIG), common channel interference avoidance cannot be enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Inter-TRP SSB staggering	INTER_TRP_SSB_STAGGER_SWITCH option of the NRDUCellMultiTrpAlgoSwitch parameter	<i>Hyper Cell Cell Combination</i>	When DL Blanking is enabled (by setting the NRDUCell.PeerCellsSlotAssignment parameter to a value other than NOT_CONFIG), inter-TRP SSB staggering cannot be enabled.
Low-frequency TDD	SSB Time Position	NRDUCell.SsbTimePos	None	When DL Blanking is enabled (by setting the NRDUCell.PeerCellsSlotAssignment parameter to a value other than NOT_CONFIG), SSB Time Position cannot be set to any of the following values: • SSB2 to SSB7 • SSB0_MOD3 to SSB2_MOD3
Low-frequency TDD	Inter-FR CA	INTER_FR_FR1TOFR2_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	When DL Blanking is enabled (by setting the NRDUCell.PeerCellsSlotAssignment parameter to a value other than NOT_CONFIG), inter-FR CA cannot be enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Smart SSB beam	SMART_SSB_BEAM_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	None	When DL Blanking is enabled (by setting the NRDUCell.PeerCellsSlotAssignment parameter to a value other than NOT_CONFIG), smart SSB beam cannot be enabled.
Low-frequency TDD	Remote interference adaptive avoidance	RIM_ADAPT_AVOID_SW option of the NRDUCellAlgoSwitch.RimAlgoSwitch parameter	<i>Remote Interference Management (Low-Frequency TDD)</i>	When the NRDUCellTimeInterface.CrossLinkInterface.Cfg parameter is set to CROSS_LINK_SEQ_S_D_TYPE1 , the RIM_ADAPT_AVOID_SW option of the NRDUCellAlgoSwitch.RimAlgoSwitch parameter must be deselected.
Low-frequency TDD	Enhanced remote interference adaptive avoidance	RIM_ADAPT_AVOID_ENH_SW option of the NRDUCellAlgoSwitch.RimAlgoSwitch parameter	<i>Remote Interference Management (Low-Frequency TDD)</i>	When the NRDUCellTimeInterface.CrossLinkInterface.Cfg parameter is set to CROSS_LINK_SEQ_S_D_TYPE1 , the RIM_ADAPT_AVOID_ENH_SW option of the NRDUCellAlgoSwitch.RimAlgoSwitch parameter must be deselected.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Co-MM	INTRA_GNB_SU_COMM_SW option of the NRDUCellCo llabServ.Mul tiCellMimoS witch parameter	<i>Co-MM (Low-Frequency TDD)</i>	Co-MM can be enabled only when the NRDUCell.PeerCe llSlotAssignment parameter is set to NOT_CONFIG .
Low-frequency TDD	Inter-cell interference avoidance	INTER_CELL_INTRF_AVOID_SW option of the NRDUCellCo llabServ.Mul tiCellMimoS witch parameter	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	When inter-cell interference avoidance is enabled, the NRDUCell.PeerCe llSlotAssignment parameter must be set to NOT_CONFIG .

11.3.2.3 Function Impacts

Some features or functions are not supported when certain slot configurations are used. For details about the slot configuration requirements of features or functions, see the cell requirements described in the corresponding feature documents.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink intra-FR inter-band CA	INTRA_FR_INTER_BAND_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	Downlink intra-FR inter-band CA can take effect only when the absolute value of the frame offset difference between the PCell and an SCell is less than or equal to 625 Ts. As such, the frame offset modified due to Slot Configuration (TDD) must meet this requirement to prevent downlink intra-FR inter-band CA from not taking effect. For details about the requirements of the downlink intra-FR inter-band CA function for the frame offset difference between the PCell and SCells, see the "Networking" section of this function in <i>Carrier Aggregation</i>.
Low-frequency TDD	Uplink intra-FR inter-band CA	INTRA_FR_INTER_BAND_UL_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	If DL Blanking is enabled in the PCell, the function of multiple timing advances (MTA) cannot be used for CA UEs, affecting uplink peak data rates of these UEs.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Clock out-of-synchronization detection	CLK_DETECT_SW option of the gNodeBParam.ClkOutofSyncDetSwitch parameter	<i>Synchronization</i>	When sequences for clock out-of-synchronization detection are being sent or checked, sending and check of cross-slot sequences are stopped.
Low-frequency TDD	Common-channel dynamic shutdown in TTI-level channel shutdown	NRDUCellPowerSaving.SsbDynShutdownSwitch parameter and the TTI_RF_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	With the DL Blanking function enabled, the maximum number of SSBs decreases, and common-channel dynamic shutdown (a sub-function of TTI-level channel shutdown) may take effect in more scenarios.

11.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

- 3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.
- DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

DL Blanking, interfering cell detection, and the following slot structures can be used only with 3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

- Slot structures SS01001202 and SS18 for the 4:1 slot configuration
- Slot structure SS518 for the single-period 8:2 slot configuration
- Slot structure SS818 for the dual-period 8:2 slot configuration

Boards

All NR-capable main control boards and NR TDD-capable baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

Cells with different frame offsets must be bound to different BBPs. The frame offsets of all cells on the same BBP must be the same.

TDD cells with different slot configurations must be bound to different BBPs. The slot configurations of all TDD cells on the same BBP must be the same.

DL Blanking and interfering cell detection require UBBPg series boards, which are the only type of baseband processing units that are compatible with these functions.

RF Modules

Setting slot configurations is required for all NR TDD-capable RF modules. The following constraints on slot configuration capabilities and NR cell configurations apply:

- Only the RRU5818, RRU5810, RRU5818w, and RRU5815 support the SS01001202 slot structure for the 4:1 slot configuration.
- NR cells served by the same RF module must use the same frame offset, slot configuration, and slot structure. When multiple cells in different NR frequency bands are served by the same RF module among the pRRU565xAB/pRRU575xAB/pRRU585xAB series modules (*x* indicates a digit, and *A* and *B* indicate different letters), AAU5665md, and AAU5368md, these cells can use different frame offset, slot configuration, and slot structure settings.
- On RF modules working in LTE+NR mode, NR cells support only the following slot configurations if LTE cells are configured in the same frequency band:
 - 8_2_DDDDDDDSUU: indicating the single-period 8:2 slot configuration
 - NR slot structure SS54 is compatible with LTE uplink-downlink subframe configuration 2 (SA2) and special subframe configuration 7 (SSP7).
 - NR slot structure SS56 is compatible with LTE uplink-downlink subframe configuration 2 (SA2) and special subframe configuration 6 (SSP6).
 - NR slot structure SS518 is compatible with LTE uplink-downlink subframe configuration 2 (SA2) and special subframe configuration 5 (SSP5).
 - 8_2_DDDSUUDDDD: indicating the dual-period 8:2 slot configuration
 - NR slot structure SS84 is compatible with LTE uplink-downlink subframe configuration 2 (SA2) and special subframe configuration 7 (SSP7).
 - NR slot structure SS86 is compatible with LTE uplink-downlink subframe configuration 2 (SA2) and special subframe configuration 6 (SSP6).

- NR slot structure SS818 is compatible with LTE uplink-downlink subframe configuration 2 (SA2) and special subframe configuration 5 (SSP5).

11.3.4 Cells

Interfering cell detection has the following cell requirements:

- **NRDUCell.HighSpeedFlag** must be set to **LOW_SPEED**.
- **NRDUCell.NrDuCellNetworkingMode** must be set to **NORMAL_CELL** or **FUSION_CELL**.
- **NRDUCellTrp.TxRxMode** must be set to one of the following values:
 - 2T2R
 - 4T4R
 - 8T8R
 - 32T32R
 - 64T64R
 - 128T128R

11.3.5 Others

None

11.4 Operation and Maintenance

11.4.1 Data Configuration

11.4.1.1 Data Preparation

- [Table 11-10](#), [Table 11-11](#), [Table 11-12](#), [Table 11-13](#), and [Table 11-14](#) describe the parameters used for function activation. This section does not describe parameters related to cell establishment.
 - [Table 11-10](#) describes the parameters that must be set in an **NRDUCell** MO to configure an NR DU cell.

Table 11-10 Parameters related to an NR DU cell

Parameter Name	Parameter ID	Setting Notes
NR DU Cell ID	NRDUCell.NrDuCellId	Set this parameter based on the network plan.
Cell ID	NRDUCell.CellId	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Duplex Mode	NRDUCell.DuplexMode	Set this parameter based on the network plan.
Frequency Band	NRDUCell.FrequencyBand	Set this parameter based on the network plan.
Subcarrier Spacing	NRDUCell.SubcarrierSpacing	Set this parameter based on the network plan.
Slot Assignment	NRDUCell.SlotAssignment	This parameter only needs to be set for the TDD mode.
Slot Structure	NRDUCell.SlotStructure	This parameter only needs to be set for the TDD mode.
Peer Cell Slot Assignment	NRDUCell.PeerCellSlotAssignment	In the case of a single slot configuration, set this parameter to NOT_CONFIG . When 4_1_DDDSU and 8_2_DDDDDDDDSUU coexist, set this parameter to 8_2_DDDDDDDDSUU or 4_1_DDDSU if the DL Blanking function is required. For details, see 11.1.2 DL Blanking .

- [Table 11-11](#) describes the parameters that must be set in a **gNodeBParam** MO to configure gNodeB parameter information.

Table 11-11 Parameters in the **gNodeBParam** MO

Parameter Name	Parameter ID	Setting Notes
Frame Offset	gNodeBParam . <i>FrameOffset</i>	If the NR slot structure and LTE subframe structure are mapped following the mapping supported by this version (for details, see 11.1 Principles), the setting notes for this parameter are as follows: • Single-period 8:2 slot configuration: The effective value of the NR frame offset is 3 ms (92160Ts) later than that of the LTE frame offset. • Dual-period 8:2 slot configuration: The effective value of the NR frame offset is aligned with that of the LTE frame offset.

- [Table 11-12](#) describes the parameters that must be set in a **gNBFreqBandConfig** MO to configure frequency band parameter information. When a multi-band cell is configured on the gNodeB, this MO is used to configure frame offsets for certain cell frequency bands.

Table 11-12 Parameters related to frequency bands

Parameter Name	Parameter ID	Setting Notes
Frequency Band	gNBFreqBandConfig . <i>FrequencyBand</i>	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Frame Offset	gNBFreqBandConfig. <i>FrameOffset</i>	If the NR slot structure and LTE subframe structure are mapped following the mapping supported by this version (for details, see 11.1 Principles), the setting notes for this parameter are as follows: • Single-period 8:2 slot configuration: The effective value of the NR frame offset is 3 ms (92160Ts) later than that of the LTE frame offset. • Dual-period 8:2 slot configuration: The effective value of the NR frame offset is aligned with that of the LTE frame offset.

- [Table 11-13](#) describes the parameters that must be set in the **NRDUCellTimeIntrf** and **NRDUCellRimConfig** MOs to configure time-domain interference and remote interference management. These MOs are required for interfering cell detection. The parameter settings must be consistent between interfering and victim cells.

Table 11-13 Parameters related to time-domain interference and remote interference management in an NR DU cell

Parameter Name	Parameter ID	Setting Notes
Cross-Link Interference Configuration	NRDUCellTimeIntrf. <i>CrossLinkIntrfCfg</i>	When this parameter is set to CROSS_LINK_SEQ_S_D_TYPE1, interfering cell detection is enabled. This function must be enabled for both interfering and victim cells.

Parameter Name	Parameter ID	Setting Notes
RIM-RS Type1 Set ID Size	NRDUCellRimConfig. <i>RimRsType1SetIdSize</i>	Set this parameter based on the number of cells to be checked. This parameter must be set to the same value for all cells where interfering cell detection is enabled.
RIM-RS Type1 Set ID	NRDUCellRimConfig. <i>RimRsType1SetId</i>	It is recommended that this parameter be set to different values for cells where interfering cell detection is enabled.
RIM-RS Frequency Candidate Number	NRDUCellRimConfig. <i>RimRsFreqCandNum</i>	Set this parameter to its recommended value. This parameter must be set to the same value for all cells where interfering cell detection is enabled.
RIM-RS Type1 Sequence Candidate Number	NRDUCellRimConfig. <i>RimRsType1SeqCandNum</i>	Set this parameter to its recommended value. This parameter must be set to the same value for all cells where interfering cell detection is enabled.
RIM-RS Type1 Scrambling ID for IS0	NRDUCellRimConfig. <i>RimRsType1ScidForIs</i> <i>0</i>	When these parameters are set to valid values for the local cell, the values must be different from each other. Each of these parameters must be set to the same value for all cells where interfering cell detection is enabled.
RIM-RS Type1 Scrambling ID for IS1	NRDUCellRimConfig. <i>RimRsType1ScidForIs</i> <i>1</i>	
RIM-RS Type1 Scrambling ID for IS2	NRDUCellRimConfig. <i>RimRsType1ScidForIs</i> <i>2</i>	
RIM-RS Type1 Scrambling ID for IS3	NRDUCellRimConfig. <i>RimRsType1ScidForIs</i> <i>3</i>	
RIM-RS Type1 Scrambling ID for IS4	NRDUCellRimConfig. <i>RimRsType1ScidForIs</i> <i>4</i>	

Parameter Name	Parameter ID	Setting Notes
RIM-RS Type1 Scrambling ID for IS5	NRDUCellRimConfig. <i>RimRsType1ScidForIs</i> 5	
RIM-RS Type1 Scrambling ID for IS6	NRDUCellRimConfig. <i>RimRsType1ScidForIs</i> 6	
RIM-RS Type1 Scrambling ID for IS7	NRDUCellRimConfig. <i>RimRsType1ScidForIs</i> 7	

 NOTE

The **NRDUCellRimConfig.*RimRsType1SetIdSize*** and **NRDUCellRimConfig.*RimRsType1SetId*** parameter settings affect the success rate and duration of cross-slot sequence detection. The parameter settings are determined based on the network plan according to the number of IDs required on the live network.

- Only interfering cell detection involves optimization configurations. [Table 11-14](#) describes the parameters used for optimization.

Table 11-14 Parameters used for optimization of interfering cell detection

Parameter Name	Parameter ID	Setting Notes
RIM-RS Scramble Timer Multiplier Factor	NRDUCellRimConfig.<i>RimRsScrTimerMulti-Factor</i>	Set this parameter to its recommended value. This parameter must be set to the same value for all cells where interfering cell detection is enabled.
RIM-RS Scramble Timer Offset	NRDUCellRimConfig.<i>RimRsScrTimerOffset</i>	Set this parameter to its recommended value. This parameter must be set to the same value for all cells where interfering cell detection is enabled.

Parameter Name	Parameter ID	Setting Notes
RIM-RS Start Frequency Offset	NRDUCellRimConfig.RimRsStartFreqOffset	Set this parameter to its recommended value. This parameter must be set to the same value for all cells where interfering cell detection is enabled and whose bandwidths and operating frequencies are the same.
RIM-RS Start Frequency RE Offset	NRDUCellRimConfig.RimRsStartFreqReOffset	Set this parameter to its recommended value. This parameter must be set to the same value for all cells where interfering cell detection is enabled and whose bandwidths and operating frequencies are the same.

11.4.1.2 Using MML Commands

Before using MML commands, refer to [11.3 Requirements](#) and complete the parameter configurations for related functions based on the impact and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Setting the slot configuration and slot structure in an NR-only network (using band n78 as an example)
MOD NRDUCELL: FrequencyBand=N78, DuplexMode=CELL_TDD, SubcarrierSpacing=30KHZ,
SlotAssignment=4_1_DDDSU, SlotStructure=SS2, NrDuCellId=0, CellId=0;
//Setting the slot configuration and slot structure in an NR-LTE dual-mode network (using band n78 as an example)
MOD NRDUCELL: FrequencyBand=N78, DuplexMode=CELL_TDD, SubcarrierSpacing=30KHZ,
SlotAssignment=8_2_DDDDDDDSUU, SlotStructure=SS54, NrDuCellId=0, CellId=0;
//Setting the frame offset using either of the following methods
//(Method 1) Setting the frame offset without specifying a frequency band
MOD GNODEBPARAM: FrameOffset=92160;
//(Method 2) Setting the frame offset of a specified frequency band
ADD GNBFREQBANDCONFIG: FrequencyBand=N78, FrameOffset=92160;
//(Optional) Enabling DL Blanking (using band n78 as an example with the slot configuration of the local cell set to 4_1_DDDSU and that of the peer cell set to 8_2_DDDDDDDSUU)
MOD NRDUCELL: FrequencyBand=N78, DuplexMode=CELL_TDD, SubcarrierSpacing=30KHZ,
```

```
SlotAssignment=4_1_DDDSU, SlotStructure=SS2, NrDuCellId=0, CellId=0,
PeerCellSlotAssignment=8_2_DDDDDDDSUU;
//(Optional) Enabling interfering cell detection (for both interfering and victim cells)
MOD NRDUCELLRIMCONFIG: NrDuCellId=0, RimRsType1SetIdSize=15, RimRsType1SetId=1023,
RimRsFreqCandNum=FREQ_NUM_1, RimRsType1SeqCandNum=SEQ_NUM_2, RimRsType1ScidForIs0=0,
RimRsType1ScidForIs1=10;
MOD NRDUCELLTIMEINTRF: NrDuCellId=0, CrossLinkIntrfCfg=CROSS_LINK_SEQ_S_D_TYPE1;
```

Optimization Command Examples

After interfering cell detection is activated, perform optimization as follows:

```
//Setting RimRsScrTimerMultiFactor, RimRsScrTimerOffset, RimRsStartFreqOffset, and
RimRsStartFreqReOffset (for both interfering and victim cells)
MOD NRDUCELLRIMCONFIG: NrDuCellId=0, RimRsStartFreqOffset=0, RimRsScrTimerMultiFactor=0,
RimRsScrTimerOffset=0, RimRsStartFreqReOffset=0;
```

Deactivation Command Examples

The following provides only command examples for deactivating the frame offset of a gNodeB, deactivating the frame offset of a frequency band of a gNodeB, disabling DL Blanking, disabling interfering cell detection, and disabling an increase in the proportion of downlink symbols in self-contained slots. You can determine whether to restore the settings of other parameters based on network conditions.

```
//(Method 1) Deactivating the frame offset of a gNodeB
MOD GNODEBPARAM: FrameOffset=0;
//(Method 2) Deactivating the frame offset of a frequency band of a gNodeB
MOD GNBFBANDCONFIG: FrequencyBand=N78, FrameOffset=4294967295;
//(Optional) Disabling DL Blanking (using band n78 as an example)
MOD NRDUCELL: FrequencyBand=N78, DuplexMode=CELL_TDD, SubcarrierSpacing=30KHZ,
SlotAssignment=4_1_DDDSU, SlotStructure=SS2, NrDuCellId=0, CellId=0,
PeerCellSlotAssignment=NOT_CONFIG;
//(Optional) Disabling interfering cell detection (for both interfering and victim cells)
MOD NRDUCELLTIMEINTRF: NrDuCellId=0, CrossLinkIntrfCfg=NOT_CONFIG;
```

11.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

11.4.2 Activation Verification

DL Blanking

- Step 1** Run the **LST NRDUCELL** command to check whether the value of **Peer Cell Slot Assignment** meets expectations. If not, change it as instructed in [11.4.1.2 Using MML Commands](#).
- Step 2** On the MAE-Access, choose **Monitor > Signaling Trace > Signaling Trace Management**. On the displayed **Signaling Trace Management** tab page, choose **Trace Type > NR > Cell Performance Monitoring > Usage of Radio Resource Monitoring**. Check the total number of used downlink RBs when packet injection

is performed at the full rate. If the total number of used downlink RBs decreases, DL Blanking has taken effect.

----End

Interfering Cell Detection

Subscribe to the external CHR event **PERIOD_CELL_CROSS_LINK_INTERF_RIM_DETECT**. If the value of **RimRsInterDbm** for an event block is not **FFFF**, interfering cell detection has taken effect. In addition, you can obtain other detection information, which is listed in [Table 11-15](#), from CHR logs. The **TrpID** field identifies a TRP of a victim cell, and the **RimRsSetID** field corresponds to an interfering cell for the TRP.

Table 11-15 Fields related to interfering cell detection and their meanings

Field	Meaning
TrpID	ID of a TRP for a cell
SymbolIndex	Detected symbol position index
RimRsSetIDNum	Detected size of the set ID of RIM-RS1
RimRsSetID	Detected set ID of RIM-RS1
RimRsInterDbm	Detected RIM-RS1 strength

11.4.3 Network Monitoring

When the base station reports ALM-26247 Configuration Failure with the alarm cause "the subframe slot setting or frame offset setting of the local eNodeB/gNodeB is inconsistent with that of adjacent eNodeBs/gNodeBs", the O&M engineer can check possibly incorrect values of base station frame offset and cell slot configuration parameters on the FMA, and replan the parameters. During the replanning, the frame offset and slot configuration reference values displayed on the FMA are for reference only.

12 Uplink Slot Configuration Optimization (High-Frequency TDD)

12.1 Principles

Commonly used slot configurations (like the 4:1 slot configuration: 4_1_DDDSU) deliver limited uplink capacity due to a small proportion of UL-only slots. Uplink Slot Configuration Optimization uses the 2:3 slot configuration to increase the proportion of UL-only slots for a higher uplink capacity.

In high-frequency TDD where the SCS is 120 kHz, the 2:3 slot configuration (2_3_DSUUU) described in [Table 12-1](#) is supported. The 2:3 slot configuration (2_3_DSUUU) comprises one DL-only slot, one self-contained slot, and three UL-only slots.

Table 12-1 Description of the 2:3 slot configuration (2_3_DSUUU) in high frequency bands

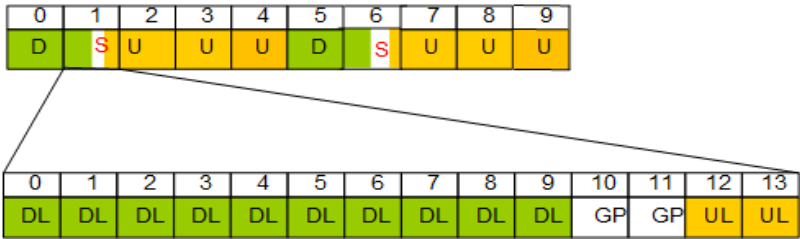
Slot Structure	GP	X (ms)	x1	x2	y2	y1
SS221	2	0.625	1	10	2	3
SS05080402	4	0.625	1	8	2	3

- The explanations are as follows:
- X: indicates the configuration period.
 - x1: indicates the number of DL-only slots.
 - y1: indicates the number of UL-only slots.
 - x2: indicates the number of downlink symbols following DL-only slots.
 - y2: indicates the number of uplink symbols before UL-only slots.
 - GP: indicates the number of guard period (GP) symbols between uplink and downlink data transmission.

Example of the 2:3 Slot Configuration (2_3_DSUUU)

A slot configuration is determined by its slot assignment and slot structure, which are specified by the **NRDUCell.SlotAssignment** and **NRDUCell.SlotStructure** parameters, respectively. [Figure 12-1](#) illustrates the slot configuration with **NRDUCell.SlotAssignment** set to **2_3_DSUUU** and **NRDUCell.SlotStructure** set to **SS221**. In this configuration, each configuration period is 0.625 ms and consists of five slots: 1 DL-only slot (**D**), 3 UL-only slots (**U**) and 1 self-contained slot (**S**). In a self-contained slot, there are 2 GP symbols (**GP**), 10 downlink symbols (**DL**), and 2 uplink symbols (**UL**).

Figure 12-1 2:3 slot configuration



12.2 Network Analysis

12.2.1 Benefits

Uplink Slot Configuration Optimization increases the number of uplink-only slots, lifting the uplink cell capacity.

This function is suitable only for scenarios where UEs are stationary. It does not ensure performance in mobile scenarios.

12.2.2 Impacts

The impacts between this function and other functions are described in [12.3.2.3 Function Impacts](#).

If intra-band or adjacent-band cells served by the gNodeBs differ in their slot configuration, interference arises between the cells, affecting the capacity and user experience of the gNodeBs. To address this issue, it is recommended that the slot configurations of neighboring cells be the same.

The 2:3 slot configuration has fewer downlink slots and therefore supports a smaller downlink cell capacity than the 4:1 slot configuration (4_1_DDDSU).

12.3 Requirements

12.3.1 Licenses

None

12.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

12.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	Downlink DMRS Maximum Length	NRDUCellPdsch.DlDmrsMaxLength	<i>Downlink Scheduling</i>	When NRDUCell.SlotAsSignment is set to 2_3_DSUUU , NRDUCellPdsch.DlDmrsMaxLength must be set to 1SYMBOL .
High-frequency TDD	CSI-RS for BM in SSB slots	SSBSLOT_CARRY_BEAM_MEAS_CSI_SW option of the NRDUCellCsirs.CsiSwitch parameter	None	If the NRDUCell.SlotAsSignment parameter is set to 2_3_DSUUU , the SSBSLOT_CARRY_BEAM_MEAS_CSI_SW option of the NRDUCellCsirs.CsiSwitch parameter must be selected.
High-frequency TDD	FR2 CA Carrier Management Policy	NRDUCellCarrierMgmt.Fr2CaCarrierMgmtPolicy	<i>Carrier Aggregation</i>	When NRDUCell.SlotAsSignment is set to 2_3_DSUUU , NRDUCellCarrierMgmt.Fr2CaCarrierMgmtPolicy must be set to USER_EXP_BASED .

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	Short PUCCH Symbol Number	NRDUCellPucch.ShortPucchSymbolNum	None	When NRDUCell.SlotAsSignment is set to 2_3_DSUUU , NRDUCellPucch.ShortPucchSymbolNum can only be set to SYMBOL1 or SYMBOL3 .
High-frequency TDD	Fixed Weight Type	NRDUCellPdsch.FixedWeightType	None	NRDUCellPdsch.FixedWeightType must be set to PMI_WEIGHT for a cell with NRDUCell.SlotAsSignment set to 2_3_DSUUU .
High-frequency TDD	PDCCH symbol number configuration	NRDUCellPdsch.OccupiedSymbolNum	<i>Channel Management</i>	When NRDUCell.SlotAsSignment is set to 2_3_DSUUU and NRCell.CellActiveState is set to CELL_ACTIVE , NRDUCellPdsch.OccupiedSymbolNum can only be set to PART_SLOT_3SYM .

12.3.2.2 Mutually Exclusive Functions

Some features or functions may not be supported when certain slot configurations are used. For details about the slot configuration requirements of features or functions, see the cell requirements described in the corresponding feature documents. The following table describes the functions that are subjected to slot configuration constraints but are not conveyed in any feature documents.

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	BFR	CONTENTION_FREE_BFR_SW option of the NRDUCellAlgoSwitch.BeamOptSwitch parameter	None	If the NRDUCell.SlotAssignment parameter is set to 2_3_DSUUU , the CONTENTION_FREE_BFR_SW option of the NRDUCellAlgoSwitch.BeamOptSwitch parameter must be deselected.
High-frequency TDD	Wide TRS beam	NRDUCellCsirs.TrsBeamType	None	If the NRDUCell.SlotAssignment parameter is set to 2_3_DSUUU , the NRDUCellCsirs.TrsBeamType parameter must be set to NARROW_BEAM .
High-frequency TDD	LNR DRX synchronization	LNR_DRX_SYNCHRONIZE_SW option of the NRDUCellUePwrSaving.NrDuCellDrxAlgoSwitch parameter	None	If the NRDUCell.SlotAssignment parameter is set to 2_3_DSUUU , the LNR_DRX_SYNCHRONIZE_SW option of the NRDUCellUePwrSaving.NrDuCellDrxAlgoSwitch parameter must be deselected.

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	Uplink MU-MIMO for mmWave large-packet UEs	MMW_LRG_PKT_UL_MU_MIMO_SW option of the NRDUCellFeatureSw.MimoFeatureSwitch parameter	None	If the NRDUCell.SlotAsSignment parameter is set to 2_3_DSUUU , the MMW_LRG_PKT_UL_MU_MIMO_SW option of the NRDUCellFeatureSw.MimoFeatureSwitch parameter must be deselected.
High-frequency TDD	PUSCH time domain enhancement	PUSCH_TIME_DOMAIN_ENH_SW option of the NRDUCellAlgoSwitch.FullChannelCovEnhSwitch parameter	None	If the NRDUCell.SlotAsSignment parameter is set to 2_3_DSUUU , the PUSCH_TIME_DOMAIN_ENH_SW option of the NRDUCellAlgoSwitch.FullChannelCovEnhSwitch parameter must be deselected.
High-frequency TDD	Intelligent beam selection	INTELLIGENT_BEAM_SELECTION_SW option of the NRDUCellBeamAlgo.BeamOptAlgoSwitch parameter	None	If the NRDUCell.SlotAsSignment parameter is set to 2_3_DSUUU , the INTELLIGENT_BEAM_SELECTION_SW option of the NRDUCellBeamAlgo.BeamOptAlgoSwitch parameter must be deselected.

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	Uplink fallback of data transmission to the MCG	UL_FALLBACK_TO_MCG_SW option of the NRCellNrDcConfig.NrDcAlgoSwitch parameter	None	If the NRDUCell.SlotAssignment parameter is set to 2_3_DSUUU , the UL_FALLBACK_TO_MCG_SW option of the NRCellNrDcConfig.NrDcAlgoSwitch parameter must be deselected.

12.3.2.3 Function Impacts

None

12.3.3 Hardware

The hardware described in this section may require compatibility between each other. For details, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

- 5900 series base stations (macro base stations)
- DBS5900 LampSite

Boards

TDD cells with different slot configurations must be bound to different BBPs. The slot configurations of all TDD cells on the same BBP must be the same.

In high-frequency TDD, a BBP can be configured with only one sector if 2:3 slot configuration is used. For details about sectors, see *Cell Management*.

The main control boards must be UMPTg or UMPTh series boards.

The BBPs must be UBBPg3, UBBPh3a, or UBBPh3b boards.

RF Modules

Macro base stations: HAAU5323, HAAU5233w, HAAU5235w, and later modules

Pole sites: HAAU5222 only

Small cell: pRRU5551JR only

If the NR carrier bandwidth is less than or equal to 400 MHz, one optical port on an RF module is used for installing optical fibers. If the NR carrier bandwidth is greater than 400 MHz, two optical ports on an RF module are required for installing optical fibers and the **NRCHAIN.77** parameter must be set to **LOADBALANCE** for the relevant RRU chain or ring.

NOTE

Among cells served by HAAU5233w and HAAU5235w, only those with the NR carrier bandwidth less than or equal to 800 MHz and **NRDUCellTrp.TxRxMode** set to **4T4R** support the 2:3 slot configuration (2_3_DSUUU).

NR cells served by the same RF module must use the same slot configuration and slot structure.

12.3.4 Cells

- The **NRDUCellTrp.TxRxMode** parameter is set to **4T4R** or **2T2R** for the cell.
- The **NRDUCell.NrDuCellNetworkingMode** parameter is set to **HYPER_CELL_COMBINE_MODE** or **NORMAL_CELL** for the cell.
- If the **NRDUCell.SubcarrierSpacing** parameter is set to **120KHZ** and the **NRDUCellPrach.PrachConfigurationIndex** parameter is set to **186** or **65535**, the **NRDUCell.CellRadius** parameter can be set to a value of **1160** or smaller.
- If the **NRDUCell.SubcarrierSpacing** parameter is set to **120KHZ** and the **NRDUCellPrach.PrachConfigurationIndex** parameter is set to **188** or **65535**, the **NRDUCell.CellRadius** parameter can be set to a value of **2410** or smaller.
- The theoretical maximum cell radius is 0.58 km when the slot structure **SS221** (two GP symbols) is used, and is 3.25 km when the slot structure **SS05080402** (four GP symbols) is used.

NOTE

- The constraint relationships between **NRDUCellPrach.PrachConfigurationIndex** set to **65535** and the **NRDUCell.CellRadius** configuration are as follows:
 - The PRACH configuration index of **65535** is automatically mapped to **186** when the cell radius is set to a value of **1160** or smaller.
 - The PRACH configuration index of **65535** is automatically mapped to **188** when the cell radius is set to a value between **1161** and **2410** (inclusive).
- The theoretical maximum cell radius indicates how far a cell can cover based on the GP length, ignoring the preamble format. (A UE farther away from the base station requires a longer time advance of signal transmission and more GP symbols. As such, the maximum cell radius can be increased by changing the slot structure.)

12.3.5 Networking

In high-frequency contiguous networking scenarios, cells must use the same slot configuration. Otherwise, interference between neighboring cells increases.

12.3.6 Others

None

12.4 Operation and Maintenance

12.4.1 Data Configuration

12.4.1.1 Data Preparation

Table 12-2 describes the parameters used for function activation. Function optimization configurations are not involved.

Table 12-2 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
NR DU Cell ID	NRDUCell.NrDuCellId	Set this parameter based on the network plan.
Cell ID	NRDUCell.CellId	Set this parameter based on the network plan.
Duplex Mode	NRDUCell.DuplexMode	Set this parameter based on the network plan.
Frequency Band	NRDUCell.FrequencyBand	Set this parameter based on the network plan.
Subcarrier Spacing	NRDUCell.SubcarrierSpacing	Set this parameter based on the network plan.
Slot Assignment	NRDUCell.SlotAssignment	This parameter only needs to be set for TDD.
Slot Structure	NRDUCell.SlotStructure	This parameter only needs to be set for TDD.

12.4.1.2 Using MML Commands

Before using MML commands, refer to **12.3 Requirements** and complete the parameter configurations for related functions based on the impact and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Setting the slot assignment and slot structure in an NR-only network (using band n258 as an example)
MOD NRDUCELL: NrDuCellId=0, DuplexMode=CELL_TDD, CellId=0, FrequencyBand=N258,
SubcarrierSpacing=120KHZ, SlotAssignment=2_3_DSUUU, SlotStructure=SS221;
```

Deactivation Command Examples

MML commands are not required for deactivating this function. You can determine whether to restore the settings of other parameters based on actual network conditions.

12.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

12.4.2 Activation Verification

- Step 1** Run the **LST NRDUCELL** command to check whether the values of **Slot Assignment** and **Slot Structure** for the NR DU cell meet expectations. If not, change them as instructed in [12.4.1.2 Using MML Commands](#).
- Step 2** Run the **DSP NRDUCELL** command to check the status of the NR DU cell. If the value of **NR DU Cell State** is **Normal**, the cell has been activated and the slot configuration has been successfully set.

----End

12.4.3 Network Monitoring

This feature ensures basic network performance and does not need to be separately monitored.

13 F-OFDM

Spectrum resources are the key to promoting the development of the wireless communications industry in the mobile communications field. F-OFDM is used to improve spectrum utilization, which is indicated by the ratio of available transmission bandwidth to channel bandwidth. The F-OFDM feature is used to achieve the maximum spectrum utilization defined in 3GPP specifications. This technology enables more spectrum resources to be used and better system performance than the OFDM does.

On the gNodeB transmitter side, the F-OFDM feature effectively controls the out-of-band leakage of transmit signals, reduces the guard band within the NR channel bandwidth, and enables more spectrum to be used for downlink transmission. On the gNodeB receiver side, the F-OFDM feature effectively controls the impact of out-of-band interference on NR, reduces the guard band within the NR channel bandwidth, and enables more spectrum to be used for uplink transmission.

Compared with the 90% spectrum utilization for LTE, F-OFDM enables higher spectrum utilization for NR, and more spectrum resources within the channel bandwidth can be used. [Table 13-1](#) lists the spectrum utilization of low frequency bands, where the channel bandwidths are for reference only. [Table 13-2](#) lists the spectrum utilization of high frequency bands. When a 20 MHz bandwidth and a 15 kHz SCS are used, the total quantity of effective spectrum resources used by NR is 1.08 MHz higher than that used by LTE. When a 100 MHz bandwidth and a 30 kHz SCS are used, the total quantity of effective spectrum resources used by NR is 8.28 MHz higher than that used by LTE.

Table 13-1 Spectrum utilization of different sub-6 GHz channel bandwidths

SCS	Channel Bandwidth	Spectrum Utilization
30 kHz	20 MHz	91.8%
30 kHz	30 MHz	93.6%
30 kHz	40 MHz	95.4%
30 kHz	50 MHz	95.76%
30 kHz	60 MHz	97.2%

SCS	Channel Bandwidth	Spectrum Utilization
30 kHz	70 MHz	97.2%
30 kHz	80 MHz	97.65%
30 kHz	90 MHz	98%
30 kHz	100 MHz	98.28%

Table 13-2 Spectrum utilization of different mmWave channel bandwidths

SCS	Channel Bandwidth	Spectrum Utilization
120 kHz	100 MHz	95.04%
120 kHz	200 MHz	95.04%

14 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

15 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

Step 1 Open the EXCEL file of performance counter reference.

Step 2 On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.

Step 3 Click **OK**. All counters related to the feature are displayed.

----End

16 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

17 Reference Documents

- 3GPP TS 38.211: "NR; Physical channels and modulation"
- 3GPP TS 38.212: "NR; Multiplexing and channel coding"
- 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
- *Cell Management*
- *Carrier Aggregation*
- *Beam Management*
- *Synchronization*
- *Hyper Cell*
- *Cell Combination*
- *Energy Conservation and Emission Reduction*
- *Interference Avoidance*
- *3D Networking Experience Improvement*
- *Remote Interference Management (Low-Frequency TDD)*
- *Co-MM (Low-Frequency TDD)*
- *Massive MIMO iBeam (Low-Frequency TDD)*
- *Technical Specifications in 3900 & 5900 Series Base Station Product Documentation*